

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks available |     |     |       |       |      |
|----------|-----|-------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6        | (a) | (i)   |  | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |     | 1   | 1     |       |      |
|          |     | (ii)  |  | <u>Nitrates</u> (levels) {peak/ are high/ increase} around {January/ winter (months)/ start of the year}(each year) (1)<br><u>Nitrate</u> levels {decrease/ are low} {between April and October/ in summer/ in July/ middle of the year} (each year) (1)                                                                                                                                                                                                                                           |                 | 2   |     | 2     |       |      |
|          |     | (iii) |  | Nitrates used by {algae/ (aquatic) plants} (1)<br>For {growth / protein} (1)<br>OR<br>In {summer/ named month} less rainfall (1)<br>less leaching of nitrates/ less run off/ less eutrophication (1)<br>OR<br>(terrestrial) plants use more nitrate in summer (1)<br>Therefore less fertiliser run off/ owtte (1)<br>2 <sup>nd</sup> mark must be linked to first                                                                                                                                  |                 | 2   |     | 2     |       |      |
|          | (b) |       |  | <b>Indicative content:</b><br><br><ul style="list-style-type: none"> <li>• Ref. fertilisers containing nitrates/ NPK</li> <li>• washed into water/ run off/ leaching</li> <li>• (the algal bloom) blocks the sunlight from aquatic plants</li> <li>• plants die</li> <li>• decompose/ break down/ decay</li> <li>• by decomposers/microorganisms/ bacteria/ fungi/ microbes</li> <li>• which use oxygen</li> <li>• in (aerobic) respiration</li> <li>• {animals/ fish} {suffocate/ die}</li> </ul> | 6               |     |     | 6     |       |      |